Fiscal Year:	FY 2010	**Task Last Updated:**	FY 08/25/2010
PI Name:	Baulch, Janet E. Ph.D.		
Project Title:	Dose rate effects on the mechanisms of space radiation induced delayed genomic instability		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jbaulch@uci.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	949.824.7396
Organization Name:	University of California, Irvine		
PI Address 1:	Department of Radiation Oncology		
PI Address 2:	Medical Sciences I, B149		
PI Web Page:			
City:	Irvine	**State:**	CA
Zip Code:	92697	**Congressional District:**	45
Comments:	NOTE: PI previously at University of Maryland; moved to UCA Irvine in summer 2013 (Ed., 11/11/13)		
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Space Radiation NNJ07ZSA001N
Start Date:	09/01/2007	**End Date:**	08/31/2011
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Change in Gaps per HRR information (Ed., 9/23/11) NOTE: End date changed to 8/31/2011 (from 8/31/2010) per S. Krenek/JSC (7/2010)		
Key Personnel Changes/Previous PI:	No Changes		
COI Name (Institution):	Morgan, William (University of Maryland)		
Grant/Contract No.:	NNX07AT42G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Among NASA's concerns for astronaut health risk is space radiation-induced carcinogenesis. At present, there is no practical way for astronauts to avoid radiation exposure, so it is important that the best possible risk estimates be made. Recent studies have demonstrated that epigenetic mechanisms play a key role in the radiation response and carcinogenesis. Epigenetic mechanisms regulate gene expression by DNA methylation, chromatin modifications and microRNA. This project tests the hypothesis that in vitro irradiation of cells alters epigenetic profiles and that these epigenetic profiles are affected radiation quality. Using proton and iron ion radiations we will investigate the delayed effects of radiation exposure in the progeny of irradiated cells multiple cell generations after the initial exposure. We will do this by measuring global, repeat element and specific locus DNA methylation. Further this project explores the possibility that DNA methylation or miRNA mediated gene silencing is involved in initiating or perpetuating radiation-induced genomic instability.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Altered DNA methylation is an epigenetic effect of exposure to space radiations that has a mechanistic role in cancer and is a measure of carcinogenic risk. We will provide a greater understanding of the potential relationship between radiation-induced genomic changes and epigenetic factors based on radiation quality. We will develop and validate high throughput biomarkers for DNA methylation. Some of the research and technology questions that this proposal addresses include the following: 1) What are the probabilities for increased carcinogenesis from space radiation exposure (protons and Fe ions, as well as x-rays)? 2) How can effective biomarkers of carcinogenic risk from space radiation be developed and validated? 3) What is the likelihood that DNA damage will increase the risk of carcinogenesis? Our data demonstrate that radiation quality affects altered DNA methylation profiles following exposure to space radiations in a radiation quality dependent nature. Changes in DNA methylation are one of the earliest and most consistent alterations associated with the carcinogenic process. Understanding the molecular mechanisms underlying these changes will provide information regarding carcinogenic processes induced by space and/or terrestrial radiations.
Task Progress:	Altered DNA methylation is an epigenetic effect of exposure to space radiations that has a mechanistic role in cancer and is a measure of carcinogenic risk. We will provide a greater understanding of the potential relationship between radiation-induced genomic changes and epigenetic factors based on radiation quality. We will develop and validate high throughput biomarkers for DNA methylation. Some of the research and technology questions that this proposal addresses include the following: 1) What are the probabilities for increased carcinogenesis from space radiation exposure (protons and Fe ions, as well as x-rays)? 2) How can effective biomarkers of carcinogenic risk from space radiation be developed and validated? 3) What is the likelihood that DNA damage will increase the risk of carcinogenesis? Our data demonstrate that radiation quality affects altered DNA methylation profiles following exposure to space radiations in a radiation quality dependent nature. Changes in DNA methylation are one of the earliest and most consistent alterations associated with the carcinogenic process. Understanding the molecular mechanisms underlying these changes will provide information regarding carcinogenic processes induced by space and/or terrestrial radiations. Year 3 accomplishments: 1. Completed DNA methylation profiles of normal and cancer cell lines exposed to 10 or 100 cGy of protons, Fe ions, or x-rays (manuscript, submitted to Radiation Research, 7/19/2010). To date, studies of radiation-induced changes in DNA methylation have focused on the low LET radiation response, using gamma- and x-rays. This study presents data demonstrating that radiation induced changes in DNA methylation are affected in a radiation quality dependent manner. This was accomplished using protons, x-rays and iron (Fe) ions. X-rays and protons are similarly low LET radiations (2 KeV/µm and 0.55 keV/µm, respectively) that differ in other ways such as energy deposition patterns or ionization density and d-ray contribution. Fe ions are a high LET particle radiation (150 keV/µm). Low LET proton and high LET Fe ions were selected for comparison to x-rays because they are the most prevalent species in space radiation. This project also tests the hypothesis for differences in the epigenetic radiation response between normal and cancer cell lines using normal human fibroblasts and colon carcinoma cells. The results demonstrate differences in basal DNA methylation profiles between normal and cancer cell lines, but similar epigenetic radiation responses that seem to be based on radiation quality. 2. Completed DNA methylation and miRNA profiles of the GM10115 CHO cell line ~16-20 population doublings following Fe ion or x-ray exposures (manuscript in preparation). This study not only further develops the picture of the effect of high and low LET radiations on DNA methylation, but also begins to delve into the potential role for miRNA in the delayed radiation response. 3. Made significant progress in developing the epigenetic profile (DNA methylation, miRNA expression and mRNA levels) of cell lines that exhibit radiation induced genomic instability. This study evaluates the link between epigenetic aberrations and cell lines that for which radiation induced chromosomal instability has been clearly documented. All DNA methylation assays and miRNA and mRNA arrays have been completed. All that remains is completing the mining of miRNA and mRNA data for pathways of interest (i.e. those involved in carcinogenic processes, DNA repair, radiation response, etc.).
Bibliography Type:	Description: (Last Updated: 05/01/2017)
Abstracts for Journals and Proceedings	Aypar U, Morgan WF, Baulch JE. "Epigenetic Alterations in Radiation Induced Genomic Instability." Presented at the 40th Annual Environmental Mutagen Society Meeting, St. Louis, MO, October 24-28, 2009. 40th Annual Environmental Mutagen Society Meeting, St. Louis, MO, October 24-28, 2009. , Oct-2009

Abstracts for Journals and Proceedings	Goetz W, Morgan M, Belliveau B, Baulch J. "Effects of High and Low LET Radiation on DNA Methylation." Presented at the 40th Annual Environmental Mutagen Society Meeting, St. Louis, MO, October 24-28, 2009. 40th Annual Environmental Mutagen Society Meeting, St. Louis, MO, October 24-28, 2009. 40th Annual Environmental Mutagen Society Meeting, St. Louis, MO, October 24-28, 2009., , Oct-2009
Abstracts for Journals and Proceedings	Goetz W, Baulch JE. "Effect of high and low LET radiation exposure on DNA methylation." 55th Annual Radiation Research Society Meeting, Savannah, GA, October 4-7, 2009. 55th Annual Radiation Research Society Meeting, Savannah, GA, October 4-7, 2009. , Nov-2009
Articles in Peer-reviewed Journals	Dziegielewski J, Goetz W, Baulch JE. "Heavy ions, radioprotectors and genomic instability: implications for human space exploration." Radiation and Environmental Biophysics, 2010 Aug;49(3):303-16. Epub 2009 Dec 25. PMID: 20035342 , Aug-2010